

RL-Tuned Synthetic Marine Radar for Sim-to-Real Transfer

IMSEL Research Group

CSCE 775: Deep Reinforcement Learning

Abstract—Machine-learning models for marine radar are often trained on synthetic data because real labeled logs are expensive to collect. A key challenge is the sim-to-real gap: models that perform well in simulation can fail on real sensor data. This proposal studies an environment-aware reinforcement learning controller that tunes simulator parameters using feedback from held-out real validation data, with environment labels such as lake, river, and coast. We compare the method against fixed randomization, Bayesian optimization, and unconditioned RL baselines, and evaluate both performance gains and sample efficiency under domain holdout and stress conditions.

I. INTRODUCTION

Marine radar is useful for obstacle detection, situational awareness, and local mapping, but robust learning requires large labeled datasets. Because collecting and labeling maritime radar logs is expensive, many pipelines rely on synthetic data. This creates a persistent sim-to-real gap: models that perform well in simulation can degrade substantially on real sensor data.

This proposal asks whether simulator parameters can be tuned automatically so synthetic training distributions better match real maritime operating regimes. We study an RL controller that updates simulator parameters to maximize held-out real-validation performance. The controller is conditioned on broad environment type (‘lake’, ‘river’, ‘coast’) with the hypothesis that context-aware tuning outperforms fixed settings and uniform randomization, especially under route and clutter shift.

The method builds on domain-randomization and adaptive sim-to-real literature [1]–[9]. Radar transfer motivation is further supported by recent radar-domain studies [10], [11] and by public real-radar datasets that expose weather and domain shift challenges [12]–[16]. Prior work often uses fixed/random perturbations, one-shot adaptation, or calibration pipelines without explicit maritime context. Our contribution is an environment-aware RL outer loop that optimizes downstream real-task reward directly (Fig. 1) and is compared against explicit capability gaps in related work (Fig. 2).

II. APPROACH

We cast simulator tuning as a bi-level optimization problem. Let $\mathcal{M}$ be the downstream perception model (e.g., MRIS-Net for ship segmentation [17]) and $\mathcal{S}(\theta, e)$ be a physically grounded radar simulator parameterized by θ and conditioned on environment type $e \in \{\text{lake, river, coast}\}$. The objective is

$$\theta^* = \arg \max_{\theta} \mathcal{R}(\mathcal{M}^*(\theta), \mathcal{D}_{real}) \quad (1)$$

subject to inner-loop training

$$\mathcal{M}^*(\theta) = \arg \min_{\omega} \mathcal{L}_{task}(\mathcal{M}_{\omega}, \mathcal{S}(\theta, e)). \quad (2)$$

The outer-loop policy $\pi(\theta_{t+1} \mid s_t)$ proposes sequential updates to simulator parameters. State s_t includes current θ , environment label e , recent validation metrics, and trajectory history. Reward is

$$r_t = \Delta \text{mIoU}_{real} - \lambda_1 \|\theta_t - \theta_{init}\|_2 - \lambda_2 \mathbb{I}(\theta \notin \Omega_{phys}), \quad (3)$$

where Ω_{phys} denotes physically plausible bounds.

The simulator models multipath and clutter using physically meaningful factors: speckle scale, attenuation, clutter floor, and multipath gain. To avoid unrealistic exploitation, each action is clipped to calibrated parameter ranges and penalized when violating physical plausibility. We use Soft Actor-Critic (SAC) for the outer loop because it supports stable learning in continuous action spaces with stochastic exploration.

The key design difference from prior adaptive randomization is that this policy is (i) sequential, (ii) environment-conditioned, and (iii) optimized on downstream real-task reward rather than simulator similarity alone.

III. EVALUATION

Evaluation uses real radar and maritime benchmarks that expose domain shift and adverse conditions, including Po-LaRIS and LaRS, with auxiliary stress testing on radar-focused datasets such as RADIATE and CARRADA [12]–[16]. We use strict splits by route/location and leave-one-environment-out testing to prevent leakage.

We compare five settings: (1) fixed simulator defaults, (2) uniform random domain randomization, (3) Bayesian optimization tuning, (4) RL tuning without environment labels, and (5) RL tuning with environment labels (proposed).

Primary outcomes are real-data task performance and sim-to-real gap reduction. Secondary outcomes include cross-domain holdout performance, unseen-route robustness, run-to-run variance, and low-SNR/high-clutter stress behavior. We also report real-validation-call efficiency to quantify adaptation cost.

Each novelty claim is tied to a direct test: (a) sequential RL adaptation versus BayRn under matched real-evaluation budget [5]; (b) environment-conditioned versus unconditioned RL; and (c) realism-constrained versus unconstrained action policies.

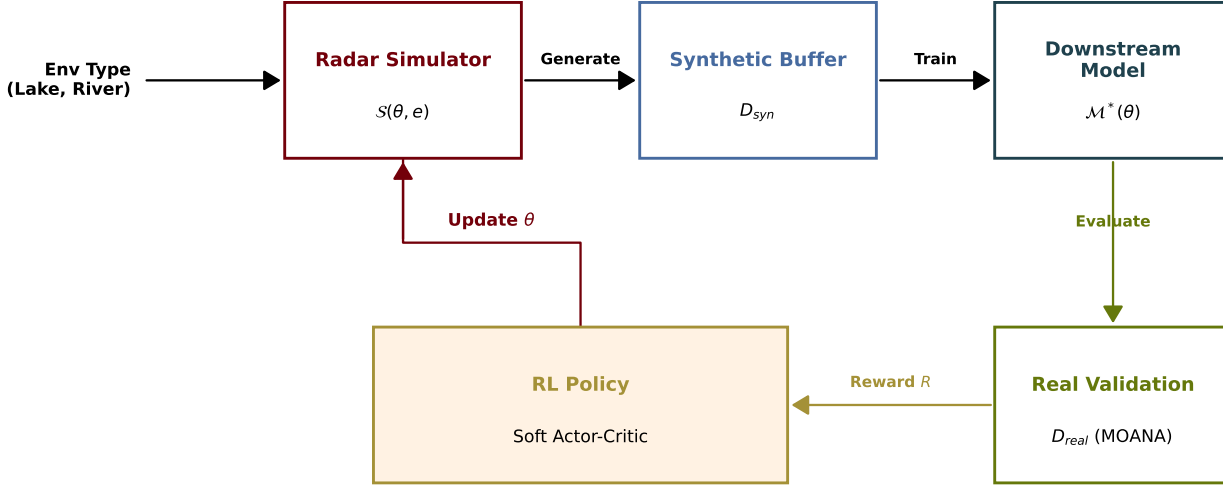


Fig. 1. Proposal 1 closed-loop method: an RL outer loop updates simulator parameters from real-validation reward, conditioned on environment type.

Method Family	Real-Feedback Adaptive	Env-Type Aware	Sequential Policy Control	Physical Bounds	Task-Reward Optimized
Fixed Domain Randomization (Tobin, Peng)	✗	✗	✗	✗	✗
SimOpt / BayesSim	✓	✗	✗	✗	✗
BayRn / ADR / CDR	✓	✗	✗	✗	✗
Radar Calibration (Digital Twin style)	✓	✗	✗	✓	✗
Proposed Method (Proposal 1)	✓	✓	✓	✓	✓

Fig. 2. Related-work capability comparison and the explicit improvement axes targeted by Proposal 1.

Published evidence already motivates this direction (Fig. 3). Real FMCW transfer can degrade toward chance-level behavior without adaptation, while calibrated adaptation improves substantially [11]. In synthetic-to-measured SAR transfer, stronger structured adaptation outperforms synthetic-only and simple-noise baselines [10]. These patterns motivate feedback-driven simulator control instead of fixed randomization.

IV. CONCLUSION

This project targets a practical problem with broad relevance beyond radar: models trained in simulation often fail when deployed in the real world. We propose an environment-aware RL controller that tunes synthetic data generation to improve real-data task performance, and we benchmark it against strong non-RL alternatives under realistic transfer stress tests. If full RL tuning is unstable, a contextual-bandit or Bayesian fallback still yields a complete and informative outcome.

REFERENCES

- [1] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, “Domain randomization for transferring deep neural networks from simulation to the real world,” *arXiv preprint arXiv:1703.06907*, 2017.
- [2] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, “Sim-to-real transfer of robotic control with dynamics randomization,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2018, pp. 1–8.
- [3] Y. Chebotar, A. Handa, V. Makoviychuk, M. Macklin, J. Issac, N. Ratliff, and D. Fox, “Closing the sim-to-real loop: Adapting simulation randomization with real world experience,” *arXiv preprint arXiv:1810.05687*, 2018.
- [4] F. Ramos, R. Possas, and D. Fox, “Bayessim: Adaptive domain randomization via probabilistic inference for robotics simulators,” *arXiv preprint arXiv:1906.01728*, 2019.
- [5] F. Muratore, M. Gienger, and J. Peters, “Data-efficient domain randomization with bayesian optimization,” *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 911–918, 2020.
- [6] OpenAI *et al.*, “Solving rubik’s cube with a robot hand via reinforcement learning and automatic domain randomization,” *arXiv preprint arXiv:1910.07113*, 2019.

Published Radar Sim-to-Real Evidence Motivating Adaptive Simulator Control

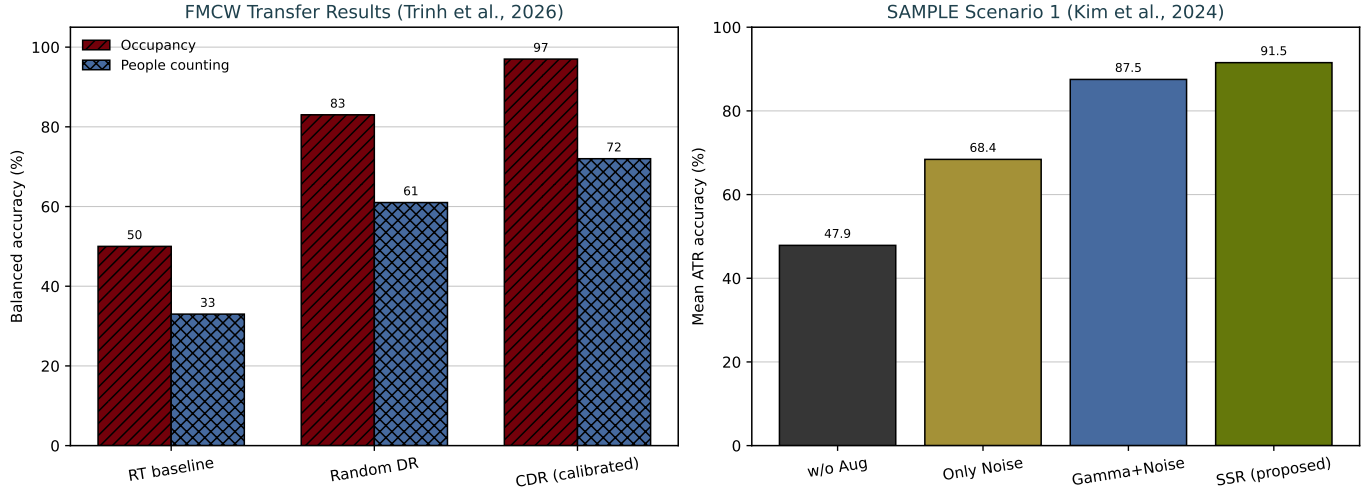


Fig. 3. Published quantitative evidence motivating Proposal 1. Left: non-adaptive and weakly adaptive baselines can fail under sim-to-real radar transfer; calibrated/adaptive methods improve substantially. Right: synthetic-only and simple noise baselines underperform on measured-data transfer compared to stronger adaptation strategies.

- [7] M. Josifovski, F. Cruz, J. Boedecker, and J. Wang, "Continual domain randomization," *arXiv preprint arXiv:2403.12193*, 2024.
- [8] J. Ho et al., "Rap: Learning to simulate robust and accurate physics for 3d manipulation," *arXiv preprint arXiv:2005.07141*, 2020.
- [9] W. Zhao et al., "Sim-to-real transfer in deep reinforcement learning for robotics: A survey," *arXiv preprint arXiv:2009.13303*, 2020.
- [10] W. Kim et al., "Soft segmented randomization: Enhancing domain generalization in sar-atr for synthetic-to-measured applications," *arXiv preprint arXiv:2409.14060*, 2024.
- [11] D. Trinh, H. Deshmukh, S. Herath, S. Waslander, and H. Trinh, "A physics-informed digital twin framework for calibrated sim-to-real fmcw radar occupancy estimation," *arXiv preprint arXiv:2601.17871*, 2026.
- [12] M. Sheeny, E. De Pellegrin, S. Mukherjee, A. Ahrabian, S. Wang, and A. Wallace, "Radiate: A radar dataset for automotive perception in bad weather," *arXiv preprint arXiv:2010.09076*, 2020.
- [13] M. Ouaknine, A. Newson, J. Rebut, F. Tupin, and P. Perez, "Carrada dataset: Camera and automotive radar with range-angle-doppler annotations," *arXiv preprint arXiv:2005.01456*, 2020.
- [14] O. Schumann, M. Hahn, J. Dickmann, and C. Wöhler, "Radarscenes: A real-world radar point cloud dataset for automotive applications," *arXiv preprint arXiv:2104.02493*, 2021.
- [15] J. Choi et al., "Polaris dataset: A maritime object detection and tracking dataset in pohang canal," *arXiv preprint arXiv:2412.06192*, 2024.
- [16] L. Zust et al., "Lars: A diverse panoptic maritime obstacle detection dataset and benchmark," *arXiv preprint arXiv:2308.09618*, 2023.
- [17] F. Ma, Z. Kang, C. Chen, J. Sun, and J. Deng, "Mrisnet: Robust ship instance segmentation in challenging marine radar environments," *Journal of Marine Science and Engineering*, vol. 12, no. 1, p. 72, 2024.